

# ***Welcome***



Dip. Informatica  
University of Pisa

- Welcome to the new students in CS
- The curricula
- ML in the curricula
- Other master degree students

***Master Degree in  
Computer Science  
(Pisa)***



# Statistics (Poll): ML 2024

**PLEASE FILL THE FOLLOWING FORM, help me to know you and help you:**



<https://forms.gle/ahwbrsQfL7xnKLj3A>

*Using the  
**UNIFI**  
account*

- How many have got the bachelor in Computer Science from **Pisa** University?  
Which is your bachelor degree?
- How many in the **AI curriculum**?
- How many in the (**Data&Know.**) **Big Data Technologies**?
- How many in the **ICT curriculum**?
- How many in the **Software curriculum**?
- How any from Master programme in **Data Science and Business Informatics**?
- How many **Digital Humanities** (Informatica Umanistica)?
- How many **Erasmus**?
- How many "**others**" and what?

# ***Master Degree in Computer Science (Pisa)***



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- Since 2017 the curricula are:
  - Artificial Intelligence (AI)
  - Data and Knowledge → (since 2021) Big Data Technologies (BD)
  - ICT Solutions Architect (ICT)
  - Software: Programming, Principles, and Technologies (SW)
- Advantages:
  - Opportunity to specialize in a field (identify your interest, professional qualification → supplementary diploma with your curriculum), ...to enjoy!
  - Methodological courses for the area at the beginning
  - Show that 2 years more of study can be useful for your future.

# Further info



## Master Degree:

- <https://www.di.unipi.it/en/education/mcs>

## Rules:

- <https://www.di.unipi.it/en/education/mcs/rules-and-resolutions>
- Instructions for each exam and
- Modality to change master programme (to the new order) or curricula or the [study plan](#)

# ML & Master degree (Pisa)

- ML is related with all the 4 curricula (see the previous discussion on the applications)
- See *characterizing* and *electives* courses in each curricula\*
- ML is *characterizing* for the AI curriculum

Artificial Intelligence



# ML & Master degree (Pisa)

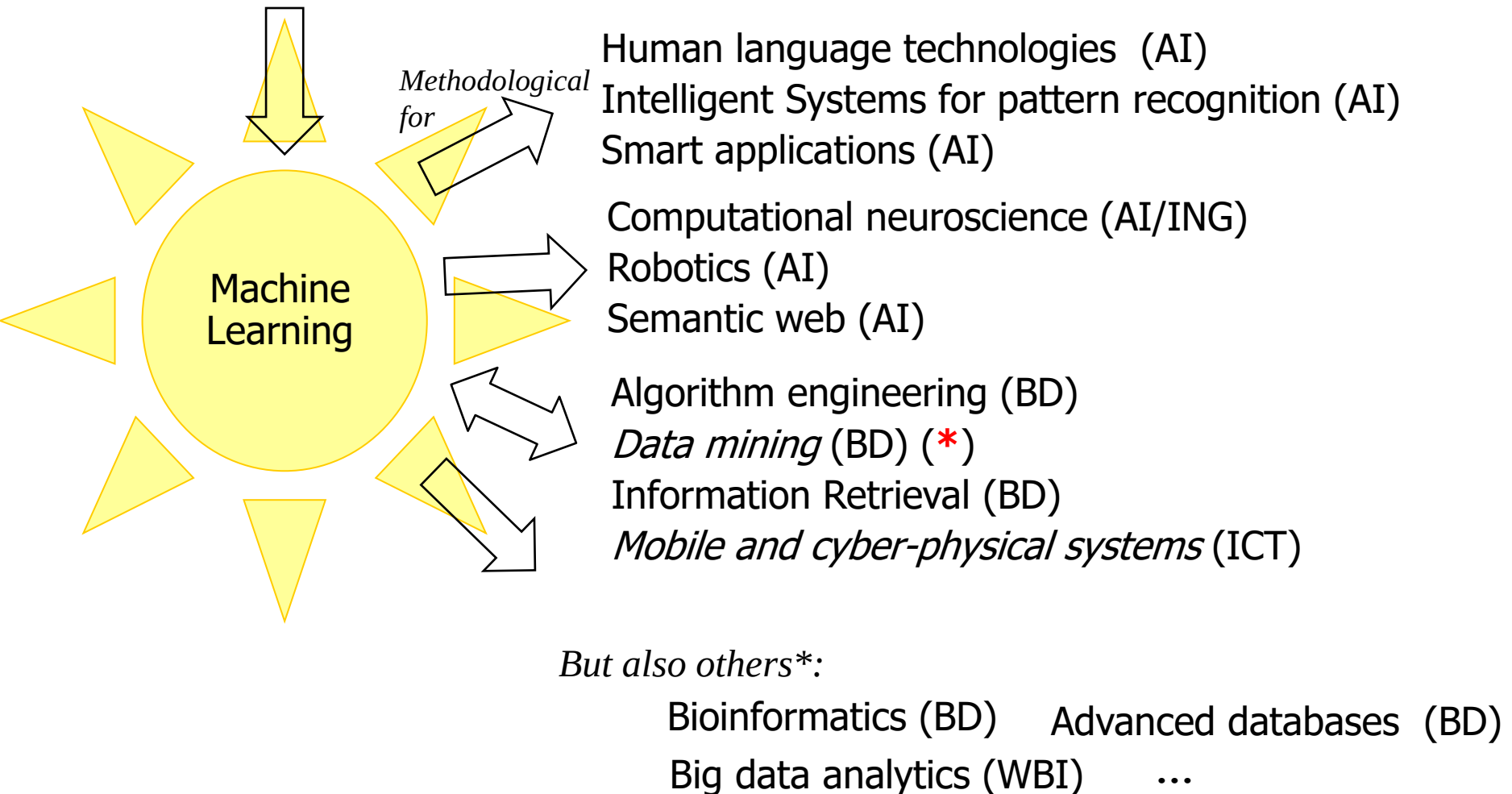
- ML is related with all the 4 curricula (see the previous discussion on the applications)
  - See *characterizing* and *electives* courses in each curricula\*
- 
- ML is *characterizing* for the AI curricula

Artificial Intelligence



# ML & Intelligent Systems area

Computational mathematics for learning and data analysis  
Parallel and distributed systems: paradigms and models



# ML & Master degree (Pisa)

- ML is related with all the 4 curricula (see the previous discussion on the applications)
  - See *characterizing* and *electives* courses in each curricula\*
- 
- ML is *characterizing* for the AI curriculum

Artificial Intelligence



# AI curriculum - Plan

- Methodological **characterizing** basis  
(to build adaptive/intelligent systems):
- Artificial intelligence fundamentals - 6 CFU
- Computational mathematics for learning and data analysis - 9 CFU
- Machine learning - 9 CFU
- Parallel and distributed systems: paradigms and models - 9 CFU

Blu: shared with other curricula

Red: as characterizing course is only in AI (but can be shared as electives)

# AI curriculum - Plan

The other **characterizing** and related fields:

- Human language technologies - 9 CFU
- Intelligent Systems for pattern recognition - ~~6 CFU~~ → 9CFU
- Smart applications - 9 CFU

**Group: AI electives** (9 CFU)

- Algorithm engineering (BD)
- Data mining (BD)
- Mobile and cyber-physical systems (ICT)

**Group: AI electives** (6 CFU) → new 2021: complete list of **11** [here](#)

- Information retrieval (BD)
- Computational neuroscience (ING)
- Social and ethical issues in computer technology
- Robotics
- Semantic web

# Further info: FAQ

## AI curriculum:

Note on Studies plan: since 2021

60 CFU **characterizing** courses

27 CFU (1 of 9 CFU and 3 of 6 CFU) from **electives**

At least 9 CFU **free choice**

Seems only **1x9 CFU + 3x6 CFU + 1 exam of 9 CFU** (free choice)

But it is also possible to choice

**1x9 CFU + 3x6 CFU + 2x6 CFU** (free choice)

i.e. 9 free choice credits covered with 2 exams of 6 CFU

Study plan link: <https://didattica.di.unipi.it/laurea-magistrale-in-informatica/piani-di-studio-3/>

# Sinergy with CM

- Take the opportunity to *follow in parallel CM and ML*
  - You get both mathematical background for learning and the ML methods , with reciprocal stimulus and continuously deepening or the underlying math/comp aspects or the modellistic effects of such choices (regardless of the order).
- In any case, for all the students, very useful opportunity for integration of the basic mathematical **background**:
  - Please see the time table by your-self
- Anyway, the course is *not* mandatory to follow ML, as many students in the past from other Master Degree or curricula without CM (we are aware of this).

# A note for students coming from IIA (Unipi BSc)



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- The first 6 lectures will be “easy” for you
- But **warning**, they are not equal to the content in IIA!
  - Take care of new parts, with different math concepts, etc.
  - We are moving from intro of the main concepts of ML through simple model examples (simple models as a means for the concepts introduction) to understanding **principles** and **models at the state-of-the-art** to solve ML tasks (models and principles as the subject).
- From lecture 5/6 ahead they are completely new.

# AA1 (320AA) $\leftrightarrow$ ML



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## ***INFO for AA1*** (old exam) $\leftrightarrow$ ***ML students***

*In summary:*

- Note: AA1 does not exist now, it is only for past students
- From AA1 to ML (you need 9 credits): I'll provide a list of topics for integrative exam, see also the symbol for the new parts:
- From ML to AA1 (you need 6 credits): **contact me as soon as possible. Anyway, I'll provide a program for the 6 credits exam.**



# Further Info?



Please ask

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